Campus Event Management System

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1. Problem Statement & Motivation

In tertiary university environments like KARE, academic departments, technical clubs, and student societies host dozens of technical symposiums, workshops, and fests each semester. Traditional manual workflows rely on disparate Google Forms, paper rosters, and uncoordinated email announcements. This causes duplicate registrations, unmanaged overbooking, lack of automated waitlisting, missed registration deadlines, and manual certificate distribution bottlenecks. The Campus Event Management System provides a single, robust full-stack portal to automate event lifecycle management, registration concurrency, student notification dispatch, and certificate access.

2. Primary Objectives

Automated Lifecycle: Allow admins to publish, update, and cancel events with seat capacities and deadlines.	**Concurrency Safety:** Prevent overbooking and double-booking using DB transactions and row-level locks.
Intelligent Notifications: Proactively alert students 24 hours prior to event registration deadlines.	**Role-Based Access:** Secure student and admin actions using stateless JSON Web Tokens (JWT) and BCrypt.

3. User Roles and Core Capabilities

User Role	Authentication & Scope	Key Functional Capabilities in Implementation
Student	Email & Register No. Role: STUDENT	• Browse published upcoming events & live seat count. • Instant 1-click registration or fair FIFO waitlist queueing. • View registered events & cancel existing registrations. • Notification dropdown with persistent read/unread count badge. • Urgent registration closing tomorrow deadline reminder banner. • View & download event participation certificates (PDF).
Campus Admin	Administrative Email Role: ADMIN	• Full CRUD: Create events, edit descriptions/instructions, cancel events. • Live registration tracking with visual capacity indicators. • Searchable participant roster per event with student department & year. • Upload verified participant completion certificates (PDF). • Automated background scheduler execution for student alerts.

4. Three-Tier System Architecture & Request Journey

Presentation Layer React 19 + TypeScript + Vite *(Port 5173 / Browser)*	➔ **HTTP / REST (JSON)** Bearer JWT in Authorization Header Vite Reverse Proxy (/api)	**Application Logic Layer** Spring Boot 3.2.5 + Spring Security *(Port 8080 / Controllers & Services)*	➔ **JDBC Driver** HikariCP Connection Pool PreparedStatements	**Database Layer** MySQL 8.0 Engine *(Port 3306 / campus_events)*

Request Lifecycle: When a student registers for an event in the React UI, a POST /api/registrations request is sent with an Authorization: Bearer <token> header. Spring Boot's JwtAuthFilter intercepts and authenticates the user. RegistrationController forwards the request to RegistrationService, which invokes RegistrationDAO. The DAO executes an ACID transaction over MySQL via a PreparedStatement, atomically locking the event row, updating seats, and recording the registration before returning structured JSON back to the browser.

Core Java & Object-Oriented Programming (OOP)

Real Implementation Analysis in the Spring Boot Backend

The backend application is written in Java 17+ adhering to object-oriented programming paradigms, clean architectural separation, and strong typing across domain models, data access objects, and service boundaries.

OOP Concept	Definition & Purpose	Where It Appears in Our Codebase	Academic & Practical Value
Classes & Objects	Blueprints representing real-world entities with state (fields) and behavior (methods).	Event.java, Student.java, Registration.java, Notification.java, Certificate.java.	Instantiates structured representations of database rows mapped into Java runtime memory.
Encapsulation	Hiding internal object state using private variables and exposing controlled getters/setters.	Notification.java: private boolean isRead; public boolean isRead() { return isRead; } public void setRead(boolean r) { this.isRead = r; }	Protects integrity of fields, supports Jackson JSON serialization aliases, and prevents unintended external mutations.
Abstraction	Hiding complex implementation mechanics and exposing a clear, clean public interface.	Controllers (e.g. EventController) only interact with high-level service contracts (EventService.updateEvent()).	Decouples web transport logic from business algorithms and raw SQL database queries.
Inheritance & Polymorphism	Reusing superclass capabilities and overriding methods for specialized runtime behavior.	JwtAuthFilter extends OncePerRequestFilter. Overrides doFilterInternal() to inject SecurityContext.	Leverages Spring Security framework infrastructure polymorphically without rewriting HTTP filter chains.
Constructors & Overloading	Specialized initialization methods for instantiating objects under different parameter states.	Event() default constructor alongside full parameterized constructor in Event.java.	Permits Jackson JSON reflection while allowing manual programmatic instantiation in DAOs.
Access Modifiers	Enforcing access boundaries using private, protected, and public.	DAOs keep helper methods (e.g. mapRow()) private while public service APIs are public.	Maintains strict package encapsulation and prevents improper invocation of internal helper routines.

Exception Handling & Robust Input Validation

Checked vs. Unchecked Exceptions: The application cleanly handles checked SQLException inside the DAO layer by logging exact SQL error codes and converting failures into appropriate business responses or unchecked RuntimeException. In EventService.java and RegistrationService.java, business rule violations throw standard IllegalArgumentException or IllegalStateException which controllers catch and translate to explicit HTTP status codes:

```
// Real validation snippet from EventService.java
if (incoming.getTotalSeats() < existing.getRegisteredCount()) {
  throw new IllegalArgumentException("Total seats cannot be less
than registered attendees (" + existing.getRegisteredCount() + ")");
}
if (incoming.getRegistrationDeadline().isAfter(incoming.getEventDat
e())) {
  throw new IllegalArgumentException("Registration deadline must be
before or equal to the event date");
}
```

HTTP Status Code Translation:
- **400 Bad Request:** Missing title, invalid dates, total seats below registered count.
- **403 Forbidden:** Student attempting admin operations (cancelling event).
- **404 Not Found:** Non-existent event, student, or certificate ID.
- **409 Conflict:** Duplicate registration attempt on the same event.
- **500 Server Error:** Unhandled database connectivity breakdown.

MySQL Database, JDBC & DAO Architecture

Relational Persistence, Concurrency Control, and Security

1. Role of MySQL & Schema Architecture

MySQL 8.0 provides durable, ACID-compliant persistence for all event activities. The database campus_events comprises 5 interrelated tables designed to prevent anomalies while maintaining strict relational integrity:

Table Name	Primary Key	Foreign Keys	Key Constraints & Integrity Enforcement
students	id (BIGINT AI)	None	uq_email, uq_register_number. Stores BCrypt hashes. Role ENUM: STUDENT / ADMIN.
events	id (BIGINT AI)	created_by ➔ students(id)	Status ENUM (DRAFT, PUBLISHED, COMPLETED, CANCELLED). Total & registered seats counter.
registrations	id (BIGINT AI)	student_id ➔ students event_id ➔ events	**UNIQUE KEY (student_id, event_id)** prevents double-booking. Status ENUM: REGISTERED, WAITLISTED, CANCELLED.
notifications	id (BIGINT AI)	student_id ➔ students event_id ➔ events	**UNIQUE KEY (student_id, event_id, notification_type)** ensures deadline alert generation is strictly idempotent.
certificates	id (BIGINT AI)	student_id ➔ students event_id ➔ events	UNIQUE KEY (student_id, event_id). UUID-prefixed stored_file_name prevents filename collision & enumeration.

2. Pure JDBC & The Data Access Object (DAO) Pattern

Rather than using heavy ORM frameworks (like Hibernate/JPA) which introduce hidden queries and reflection overhead, this project uses **pure JDBC with the DAO pattern**. This ensures transparent, highly optimized, and maintainable SQL execution:

JDBC Core Components:
• **DataSource (HikariCP):** Manages a pool of reusable connections to MySQL.
• **Connection:** Represents active physical session with the database.
• **PreparedStatement:** Pre-compiled SQL template with parameterized ? placeholders.
• **ResultSet:** Cursor-based tabular data returned from executing SQL queries.

Why Pure JDBC & DAO?
• **SQL Injection Prevention:** Parameters passed via pstmt.setString() are treated strictly as data literals, never executable SQL commands.
• **Separation of Concerns:** Controllers handle HTTP, Services govern business rules, DAOs isolate persistence.
• **Full Transaction Control:** Direct command over commit/rollback semantics.

3. Concurrency Protection & Transaction Management

Race Condition Prevention: When hundreds of students attempt to register for a limited 50-seat workshop simultaneously, naive systems cause negative seat balances (overbooking). In RegistrationDAO.java, registration is executed inside an atomic transaction:
1. connection.setAutoCommit(false); initiates transaction boundary.
2. SELECT total_seats, registered_count FROM events WHERE id = ? FOR UPDATE; locks the target event row in MySQL.
3. If registered_count < total_seats, the student is granted REGISTERED status and seat count increments.
4. If seats are exhausted, the student is automatically designated WAITLISTED without incrementing seat count.
5. connection.commit(); finalizes state. If any exception occurs, connection.rollback(); restores consistency.

Data Structures & Algorithms (DSA) in Practice

Applied Algorithms and Collection Framework Usage

1. Implemented Algorithms: MergeSort ($O(n \log n)$) in EventSorter

Algorithm Choice: To sort campus events chronologically by event date, EventSorter.java implements a pure Divide-and-Conquer **MergeSort** algorithm. Unlike Quicksort (which degenerates to $O(n^2)$ under worst-case sorted input) or Bubble/Insertion Sort ($O(n^2)$), MergeSort guarantees predictable $O(n \log n)$ time complexity and preserves stability.

Divide Step: The event list is recursively halved until sublists of size 1 are reached (trivially sorted).	**Conquer & Merge:** Sublists are merged by comparing eventDate timestamps, assembling a fully ordered list.	**Complexity:** • Time: **$O(n \log n)$** (Best, Avg, Worst) • Space: **$O(n)$** (Auxiliary buffer)

2. Java Collection Framework Implementation Matrix

The project leverages specific interfaces and concrete implementations from java.util based on algorithmic performance:

Collection / Interface	Concrete Class	Where Used in Application	Time Complexity & Justification
List<E>	ArrayList<Event> ArrayList<Registration>	DAOs for fetching event lists, user registrations, and participant tables.	**$O(1)$ random access** for indexing; optimal memory locality when transmitting query result rows.
Queue<E>	LinkedList<Long> inside WaitlistQueue.java	Waitlist processing module to track waitlisted student IDs per event.	**$O(1)$ enqueue & dequeue**; enforces fair First-In, First-Out (FIFO) promotion when a seat frees up.
Map<K,V>	HashMap<String, Object>	Controllers (e.g. AuthController) for JSON key-value response payloads.	**$O(1)$ average insertion & retrieval** of token, role, and message attributes.
Set<E>	HashSet<Long>	NotificationService to track processed event and student IDs in memory.	**$O(1)$ existence checks**; guarantees no duplicate alert generation during active batch cycles.

3. Future Algorithmic Enhancements

PriorityQueue (Min-Heap): To dynamically order imminent registration deadlines in $O(\log n)$ insertion instead of periodic cron batch scanning.	**Trie (Prefix Tree):** For real-time, sub-millisecond autocomplete search over hundreds of symposium event titles in the frontend header.

Full-Stack Tech Stack & Functional Workflows

Frontend Technologies, Authentication, and System Lifecycles

1. Technology Stack & Architectural Roles

Technology	Role in System	Technical Advantage & Why Chosen
React 19	Frontend UI View Layer	Declarative component-based architecture; dynamic virtual DOM re-renders without full-page reloads.
TypeScript	Frontend Type Safety	Enforces strict compile-time types (ApiEvent, ApiRegistration), eliminating runtime undefined errors.
Vite v8.3	Bundler & Dev Server	Sub-second hot-module replacement (HMR); fast production bundling with proxy configuration to port 8080.
Spring Boot 3.2	Enterprise REST API	Embedded Apache Tomcat web server; auto-configured beans, dependency injection, and centralized routing.
JWT & BCrypt	Stateless Security	BCrypt (cost 12) prevents hash rainbow table attacks. HS256 JWT eliminates stateful HTTP session memory.
Maven Wrapper	Build Orchestration	mvnw.cmd guarantees exact Maven 3.9.6 execution without requiring manual software installation by examiners.

2. End-to-End Student Workflow

Step 1: Auth Sign in with email & register number. Token cached in localStorage.	→	**Step 2: Browse** Explore upcoming events. Filter by department or search titles.	→	**Step 3: Register** 1-click registration. Instant seat allocation or auto-waitlist.	→	**Step 4: Engage** View registrations, deadline notices, & download certificates.

3. End-to-End Administrator Workflow

Step 1: Admin Auth Sign in with admin credentials. Granted ROLE_ADMIN.	→	**Step 2: Event CRUD** Create symposiums, modify seat limits, or cancel events safely.	→	**Step 3: Audit Rosters** Inspect attendee lists, student registration IDs, and departments.	→	**Step 4: Certificates** Upload verified participant PDF certificates to server.

Five-Member Team Work Breakdown Structure

Proposed Distribution of Engineering Responsibilities

Note for Presentation: The table below represents a balanced, modular responsibility breakdown proposed for a five-member engineering team. The names 'Team Member 1' through 'Team Member 5' are placeholders that should be customized to reflect each student's specific contributions during the academic term.

Team Member	Primary Assigned Domain	Key Modules & Engineering Deliverables
Team Member 1 *(Frontend & UI/UX)*	React 19 & TypeScript Presentation Layer	• Designed responsive university layout with navy/blue institutional aesthetic. • Built LoginPage with role toggles and demo shortcuts. • Developed StudentDashboard, EventsPage, and EventDetailPage. • Implemented interactive NotificationBell dropdown with real-time badge updates.
Team Member 2 *(Backend & Core Java)*	Spring Boot Application Layer & Security	• Architected RESTful controllers (AuthController, EventController, etc.). • Implemented service orchestration layer (AuthService, EventService). • Configured Spring Security with stateless JwtAuthFilter and BCrypt password encryption. • Enforced role-based authorization rules separating STUDENT and ADMIN endpoints.
Team Member 3 *(Database & JDBC/DAO)*	MySQL Schema & Data Access Layer	• Authored schema.sql DDL defining relational tables, foreign keys, and indexes. • Implemented pure JDBC Data Access Objects (StudentDAO, EventDAO, RegistrationDAO). • Structured PreparedStatement queries to prevent SQL injection vulnerabilities. • Designed ACID transactions with row-level locking (SELECT FOR UPDATE) to eliminate overbooking.
Team Member 4 *(DSA & Collections)*	Algorithms & Background Modules	• Implemented EventSorter.java with Divide-and-Conquer **MergeSort** ($O(n \log n)$). • Built WaitlistQueue.java using FIFO Java Collections for queue promotion. • Engineered DeadlineNotificationScheduler for 24-hour advance deadline alerts. • Ensured idempotent notification insertion using composite unique database keys.
Team Member 5 *(Testing & QA)*	Integration Testing, Build & Documentation	• Authored JUnit 5 automated test suites: EventCancellationTest and NotificationServiceTest. • Configured Maven Wrapper (mvnw.cmd) ensuring reproducible execution across laptops. • Conducted end-to-end regression testing on event cancellation and mark-as-read workflows. • Authored comprehensive evaluator README.md and managed GitHub version control releases.

Collaborative Engineering & Peer Code Review

While specific domains were assigned to individual leads, all five members participated collectively in daily standups, cross-module code reviews, integration testing, database schema refinement, and bug-fixing sessions. Key achievements like resolving the Jackson boolean serialization bug on notification unread badges and ensuring HTTP 403 authorization compliance on event cancellations were collaborative triumphs.

Testing, Viva Revision Guide & Academic Conclusion

Quality Assurance, Common Viva Questions, and Final Summary

1. Quality Assurance: Automated vs. Manual Testing Matrix

Test Type	Target Component	Test Case Scenarios & Verified Assertions	Result
Automated (JUnit 5)	EventCancellationTest	Admin can cancel; Student cancellation rejected with 403 Forbidden; unauthenticated rejected; DB status changes to CANCELLED.	**PASSED** (6/6 tests)
Automated (JUnit 5)	NotificationServiceTest	Calculates registration deadlines expiring tomorrow; creates in-app alerts; verifies idempotency on repeated execution.	**PASSED** (4/4 tests)
Automated (Vite/TS)	Frontend Type Check & Build	npm run build verifies TypeScript syntax, API client models, and packages optimized static chunks.	**PASSED** (0 errors)
Manual Verification	End-to-End User Journeys	Verified student login, event registration, waitlist queuing, mark-read decrementing, and PDF certificate downloads.	**VERIFIED**

2. Top 10 Viva Revision Questions & Concise Model Answers

Q1: Why did you use Spring Boot with pure JDBC instead of Hibernate / JPA?
Pure JDBC with PreparedStatement provides full transparency, predictable execution, and zero ORM caching bugs. It meets academic requirements to demonstrate SQL mastery and transaction management.

Q2: How does the system prevent SQL Injection attacks?
By strictly using PreparedStatement with parameterized queries (? placeholders). Values are sent separately from the SQL syntax and treated as literal values, never executable SQL.

Q3: How is concurrency and overbooking prevented during high-traffic registrations?
In RegistrationDAO, registration runs inside a transaction with SELECT ... FOR UPDATE. This places a row-level lock on the event until seats are checked and incremented.

Q4: What is the purpose of the DAO pattern?
Data Access Object isolates database persistence code from business logic. Services call methods like eventDAO.findById() without caring about SQL syntax or table structures.

Q5: How does the waitlist queue work algorithmically?
When an event is full, registrations are assigned WAITLISTED. WaitlistQueue.java models this as a First-In, First-Out (FIFO) queue ($O(1)$ operations) promoting the earliest registrant.

Q6: Why did you choose MergeSort over QuickSort for EventSorter?
MergeSort guarantees $O(n \log n)$ time complexity in all cases (best, average, and worst). QuickSort can degrade to $O(n^2)$ if pivot selection encounters already sorted event lists.

Q7: How is user authentication handled securely?
Passwords are encrypted with BCrypt (cost factor 12). On login, the backend issues an HS256-signed JWT token containing user ID and role, verified on every request by JwtAuthFilter.

Q8: How does the registration deadline reminder system work?
A Spring @Scheduled task runs daily at 8:00 AM, scanning events with deadlines in the next 24 hours and creating notifications. Database unique constraints prevent duplicate alerts.

Q9: What is the difference between List, Set, and Map in your project?
List stores ordered rows with duplicates (events); Set stores unique elements ($O(1)$ duplicate check); Map stores key-value pairs for JSON responses ($O(1)$ lookup).

Q10: What future improvements could be added?
Implementing automated email dispatch via JavaMailSender, integrating payment gateways for ticketed events, and introducing Redis caching for high-speed read operations.

3. Academic Conclusion

The Campus Event Management System delivers a complete, production-ready full-stack solution tailored for university operations. It successfully demonstrates core computer science competencies: object-oriented software engineering, relational database design, concurrency protection, algorithmic sorting, and secure RESTful web architecture. The project is verified, fully functional, and ready for evaluation.